Elicit vs. Teach, Mechanistically: Results on the TinyStories-1B Twins

MARS V — Mechanistic Understanding of Elicitation vs. Teaching

Working results document, September 2026

Setup

Two identical TinyStories-1B twins (same architecture, same pretraining on children's stories, zero arithmetic) are fine-tuned on the same target task — signed 1–4-digit addition/subtraction in natural language, `What is the difference between 8 and 5784?` → `-5776` — with one byte-held protocol (LoRA r=512, α=32, lr 3.53×10^{-4}, batch 128, seed 316, eps/k convergence stop, frozen D_algo_bare, $n = 10^2 \ldots 10^6$). The arms differ *only* in the parent checkpoint:

- **Teach twin**: the blank base (`evt-ts1b-base`). The capability is absent.
- **Elicit twin**: **TS1B-latent** (`evt-ts1b-op-bridge-mix`) — the blank base plus (i) a symbol-only arithmetic install (`23 + 45 = 68`, 1M examples, full FT) and (ii) an answer-free word↔symbol binding dose (rewriting `What is the sum of 621 and 5068?` ↔ `621 + 5068`, rehearsed 1:1 with already-seen symbol rows). Every installation is answer-free and leakage-controlled (Table 1). The capability is latent.

Measurement vocabulary. *Circuit* = top-32 of 528 nodes (512 attention heads + 16 MLP outputs) by attribution score (gradient of the clean/corrupt logit-diff × activation delta, 128–256 pairs); overlaps are Jaccard@32 against a chance floor of 0.031 and a per-map split-half ceiling. *Weight write* = ΔW = scaling $\cdot\, BA$ per LoRA module (exact; singular values via thin QR); total write $\|\Delta W\|_F$, relative write $\|\Delta W\|/\|W\|$, travel $\|\Delta W_t\|_F$ and speed $\|\Delta W_t - \Delta W_{t-1}\|_F/\Delta t$ from adapter snapshots. *Gradient strength* = the pre-clip global gradient norm $\|g_t\|$ logged at every update (runs' `gradstats.jsonl`); cumulative gradient mass $\sum_t \|g_t\|$.

Construction step / check	value	control
symbol install: EM on `23 + 45 =`	0.67–0.73	blank: 0.00
corpus census: "sum" in 439M words	21×	"equals" 17, "minus" 57
binding dose: NL→op rewrite exact	0.484	blank + dose: 0.000 EM
sum-only dose on *difference* questions	writes "+"	per-word binding
direct NL exact match of the parent	**0.000**	= blank twin

Table 1: The latent parent is behaviorally identical to the blank twin on the target task. Everything below is about what differs inside.

Results

R1 The learning-curve signature flips.

Metric. Prequential EDL per label token (nats) and held-out exact match, as a function of dataset size n, identical protocol, parent varied.

Result. Figure 1, Table 2. Elicit: EDL strictly monotone, $4.53 \to 0.092$; EM 0.54 at n=3,162, 0.98 at 1M. Teach: EDL down–up–down with the hump peaking near n=215K; EM 0.093 at 1M. EM-0.5 threshold: $\sim 3 \times 10^3$ vs $> 10^6$ ($> 300\times$). A third arm (blank + format-only dose) keeps its hump.

Elicit vs. teach. Eliciting spends every example connecting an existing capability (diminishing cost from the first rung); teaching must build the capability, producing increasing returns mid-run. The flip is the paper's causal-intervention claim reproduced on one substrate, with a parent whose latency is itemised rather than assumed.

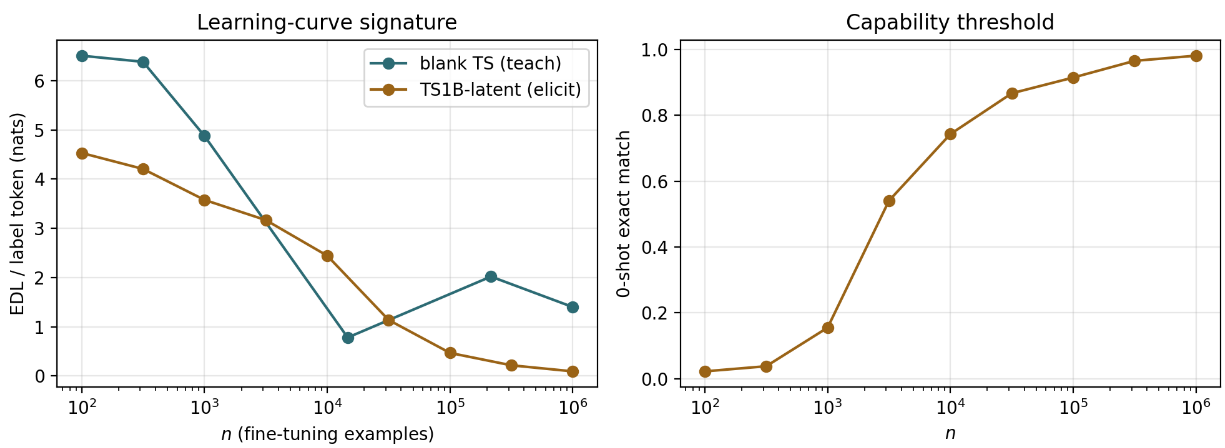

Figure 1: EDL/token (left) and 0-shot EM (right) vs n; teal = teach twin, gold = elicit twin.

n	elicit EDL/tok	elicit EM	teach EDL/tok	teach EM
100	4.533	0.023	6.511	0.007
316	4.208	0.038	6.387	0.033
1,000	3.579	0.155	∼4.9	∼0
3,162	3.168	**0.541**	↓	∼0
10,000	2.444	0.743	min ∼0.8	∼0
31,623	1.137	0.867	↑	∼0
100,000	0.467	0.915	↑	∼0
316,228	0.215	0.966	peak ∼2.0 @215K	∼0
1,000,000	**0.092**	**0.981**	1.40	0.093

Table 2: Per-rung values (EDL nats/label token; EM on 1,024 held-out problems).

R2 The elicited circuit is the parent's installed circuit; the taught circuit matches nothing upstream.

Metric. Jaccard@32 between attribution circuits (chance 0.031; ceiling = split-half self-agreement of the map, 0.684 for the elicited child).

Result. Table 3. Parent (installed engine) ↔ elicited child: 0.455–0.524 across eval surfaces (≈70% of ceiling), top-4 nodes identical and identically ordered (`mlp:15,14,13,12`). Blank parent: no measurable circuit (logit-diff −0.33 even 16-shot; its map is noise), so the taught child has nothing to match.

Elicit vs. teach. Elicitation re-weights machinery that was already there; teaching constructs machinery where there was none. "Reuse" is measured, not inferred.

Comparison	J@32	chance	ceiling
installed engine ↔ elicited child (op / bridge surface)	0.455–0.488 / 0.524	0.031	0.684
blank parent ↔ taught child	undefined (parent map is noise)	–	–
elicited child @3K ↔ @1M	0.391	0.031	0.684
elicited child @1M ↔ taught child @1M	0.231 (Spearman −0.11)	0.031	0.684 / 0.561

Table 3: Circuit-membership overlaps. All maps pass the performing-regime guard.

R3 Same task, same substrate, different machine.

Metric. Layer distribution of the top-16 attribution nodes; J@32 and score Spearman between the two children.

Result. Figure 2. Elicited child: mid/late MLP backbone (layers 5–15) plus the layer-7 attention trio — the installed engine's own profile. Taught child: eight layer-0 attention heads over raw digit tokens feeding a few late MLPs. Overlap 0.231, score correlation −0.11.

Elicit vs. teach. Because both children share the identical pretrained substrate, the mechanism difference is caused by the learning regime alone — the cross-model version of this finding had a pretraining confound; this one does not.

R4 The elicited circuit exists before training; the taught circuit is built mid-run — through the EDL hump.

Metric. Jaccard@32 of each training snapshot's circuit against the run's final circuit.

Result. Figure 3. Elicit: 0.488 at step 1 (16× chance), a brief interface-wiring transient (dip to ∼0.42 with a strong map at step 154), locked from ∼step 1.5K of 11.5K (0.561; 82% of ceiling, 97% of the J@64 ceiling by 8.4K). Teach: noise floor (≈0.22–0.28) until ∼step 1.3K, then 0.27 → 0.44 over steps 1.3K–5.4K — the region where its EDL peaks.

Elicit vs. teach. Elicitation finds and wires; teaching constructs. The elicited circuit finishes stabilising before the taught one begins to exist, and the teaching hump is literally the circuit being assembled.

R5 The words reach the engine only when the binding is installed.

Metric. Cross-format activation matching: per node, mean cosine between the activation on the NL rendering and on the symbol rendering of the *same* problem, minus the mismatched-pair baseline ("does `What is the sum of 621 and 5068?` drive the node into the same problem-specific state as `621 + 5068 = ?`").

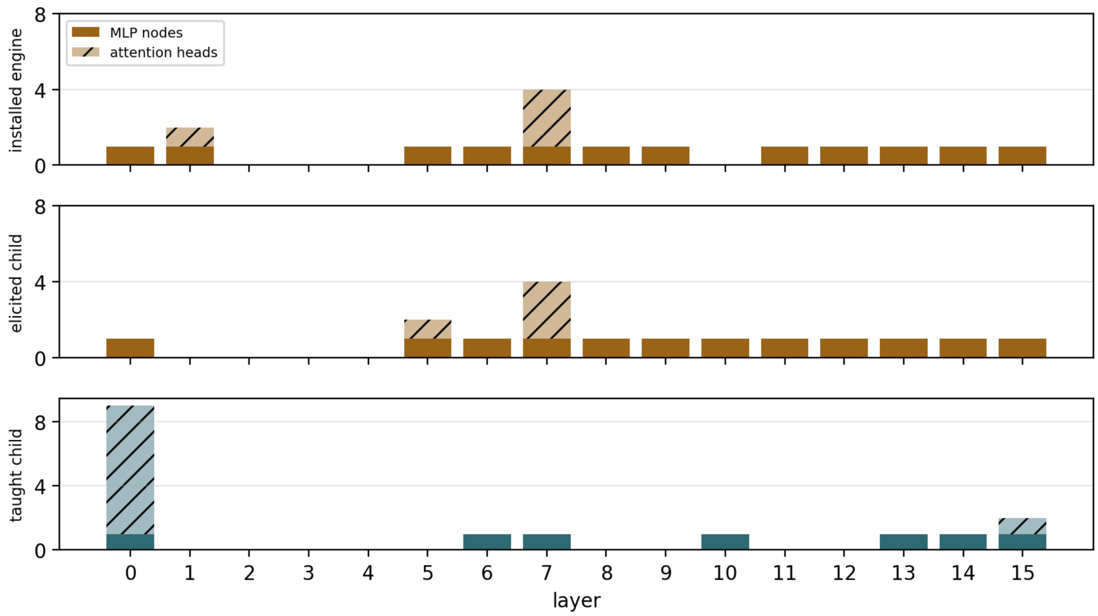

Figure 2: Top-16 node layer histograms (solid = MLP, hatched = attention) for the installed engine, the elicited child, and the taught child.

Result. Engine without binding: MLP index ≈ 0.00–0.01 at every layer. TS1B-latent (binding installed): +0.07–0.13 across layers 7–15, ∼10×. Attention-head indices are confounded by shared digit tokens and are excluded.

Elicit vs. teach. This is what "latent" means internally: before any target training, NL input already activates the arithmetic MLPs of the elicit parent, while the teach parent has no arithmetic MLPs to activate. It also shows the binding dose did what it claimed — installed an access route, not the task.

R6 The latent capability can be reached with frozen weights; the absent one cannot.

Metric. Exact match on the target task under zero-training interventions: (a) self-chain — the model rewrites the question itself, then computes its own rewrite; (b) donor patching — the fine-tuned child's activations at the 32 circuit nodes written into the parent (mean vector, or per-prompt states); random-node controls.

Result. Figure 4, Table 4. Elicit parent: direct 0.000 → self-chain **0.328** (= 0.484 × 0.668; the composition law parse×compute held on four model variants); per-prompt states 0.125 (random nodes: 0.000); mean vector: format only. Teach side: blank + same doses 0.000; blank + chain 0.000 (a parser without an engine); taught donor into blank twin 0.000 in every condition.

Elicit vs. teach. Elicit = one composition step or one state-patch away from the target; teach = nothing underneath to switch on. The teach-side nulls double as the leakage control for the whole construction.

R7 Teaching writes far more into the weights, and buys far less per unit written.

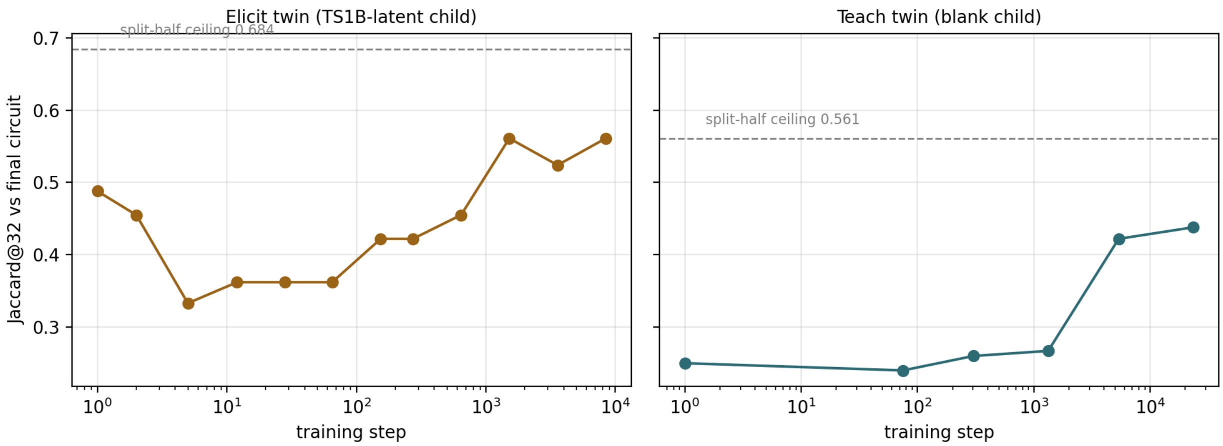

Figure 3: Circuit formation, elicit (left) vs teach (right); dashed = split-half ceiling.

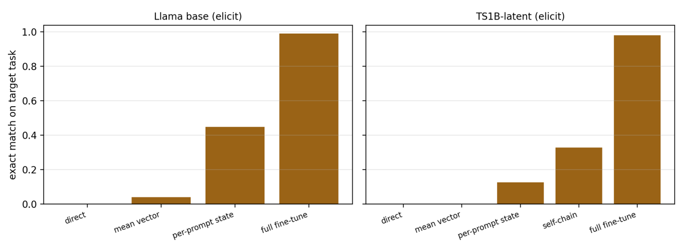

Figure 4: Intervention ladder on the elicit parent (weights frozen except the last bar).

Metric. Relative write $\|\Delta W\|/\|W\|$ (exact from LoRA factors); total travel and per-step speed from adapter snapshots; capability per unit travel = final EM / total travel.

Result. Table 5, Figure 5. Relative write 0.212 (teach) vs 0.095 (elicit), 2.2×; total travel 457 vs 120, 3.8×, over 2× as many steps (23.5K vs 11.5K); capability per unit travel 2.0×10^{-4} vs 8.1×10^{-3}, ∼40×.

Elicit vs. teach. Eliciting connects what exists; teaching writes what does not. The temporal *shape* of writing (burst then decay) is shared by both arms — an optimisation signature — so the discrimination is in amplitude, duration, and efficiency, not timing.

R8 Teaching's gradient grows through training; elicitation's decays.

Metric. Pre-clip global gradient norm at every update, from the runs' `gradstats` logs: peak, mean over first 1% / middle / last 10% of steps, cumulative gradient mass $\sum_t \|g_t\|$, and its split across weight classes.

Result. Table 6, Figure 6. Both arms start at the same scale (first-1% mean 0.18 vs 0.17). Elicit then decays ×3.3 to a floor of 0.056 (peak 0.44 at step 96). Teach *grows* ∼40× to 6.8 (peak 17.4 at step 23,657 — still rising when the run stopped). Cumulative gradient mass 783 vs 116,877 (∼150×). Across n the arms are indistinguishable up to 10K and diverge from 31.6K on (mass ratio 7× → 55× → 95× → 150×). Teach's gradient mass migrates out of the MLPs (share 0.47 → 0.01) into attention (VO 0.87).

Patient + intervention (k=32 circuit nodes)	condition	format	EM
TS1B-latent + child mean shift	circuit nodes	0.164	0.000
TS1B-latent + child per-prompt states	circuit nodes	0.359	**0.125**
	random-32 nodes	0.051	0.000
TS1B-latent self-chain (no donor)	own rewrite + "= "	–	**0.328**
blank + binding dose, self-chain	parser without engine	–	0.000
blank + taught donor	every condition	0.000	0.000

Table 4: Zero-training interventions, TinyStories twins.

	elicit (TS1B-latent)	teach (blank)
relative write $\|\Delta W\|/\|W\|$ (1M)	0.095	0.212
steps to convergence	11,500	23,496
peak / final writing speed (per step)	0.218 / 0.018	0.306 / 0.022
total weight travel $\|\Delta W\|_F$	120	457
final exact match	0.981	0.093
capability per unit travel	8.1×10^{-3}	2.0×10^{-4}

Table 5: Weight-space write, elicit vs teach.

Elicit vs. teach. The clearest dynamical discriminator: eliciting converges and its gradient dies; teaching keeps generating gradient pressure that grows with the structure being built, and that pressure is aimed at attention — the layer-0 army of R3 being written. It also explains R7: AdamW's $\sqrt{v}$-normalisation compressed a 150× gradient-mass difference into a 3.8× travel difference with a shared temporal shape, so raw gradient norm — not update size — is where the regimes separate.

1M fine-tune	elicit	teach
peak grad norm (step)	0.441 (96)	17.4 (23,657)
mean norm: first 1% / middle / last 10%	0.183 / 0.060 / 0.056	0.169 / 5.21 / 6.80
first 1% ÷ last 10%	×3.3 (decays)	×0.02 (grows ∼40×)
cumulative gradient mass $\sum_t \|g_t\|$	783	116,877
gradient mass per step	0.068	4.68
mass share QK / VO / MLP	.18 / .55 / .27	.12 / .87 / .01

Table 6: Raw gradient strength, 1M fine-tunes (pre-clip global norm at every update).

R9 The regime difference is total: the two children differ at both node and edge level.

Metric. $\Delta S = 1 -$ Jaccard at node level (@32) and edge level (@256, node→block edge attribution).

Result. Elicited @1M vs taught @1M: $\Delta S_{\text{node}} = 0.769$, $\Delta S_{\text{edge}} = 0.739$ (ratio 0.96), both far above the noise floors (0.32 / 0.61).

Elicit vs. teach. When the mechanism differs, it differs everywhere — membership and wiring alike — consistent with R3: not a re-routed version of one machine but two machines.

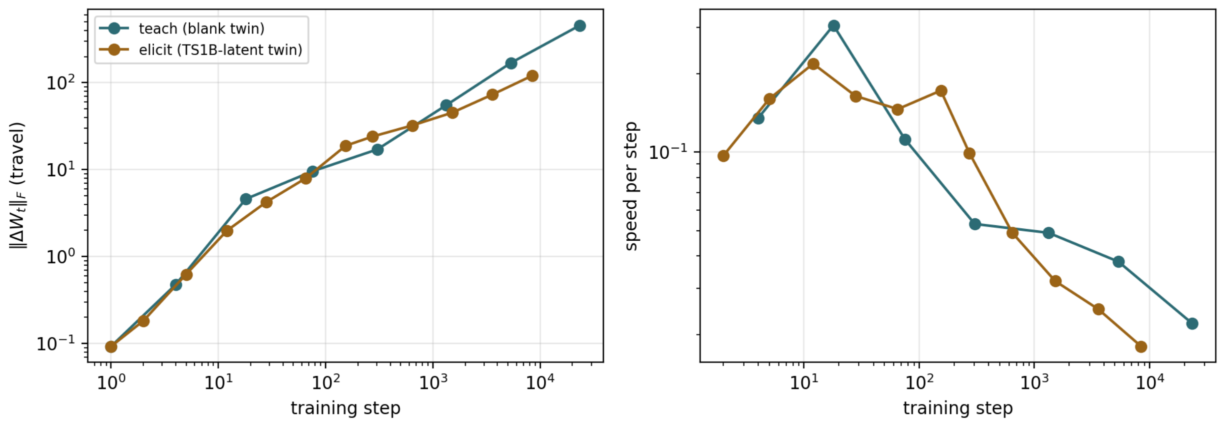

Figure 5: Weight travel (left) and speed (right) from adapter snapshots; profiles coincide, amplitudes do not.

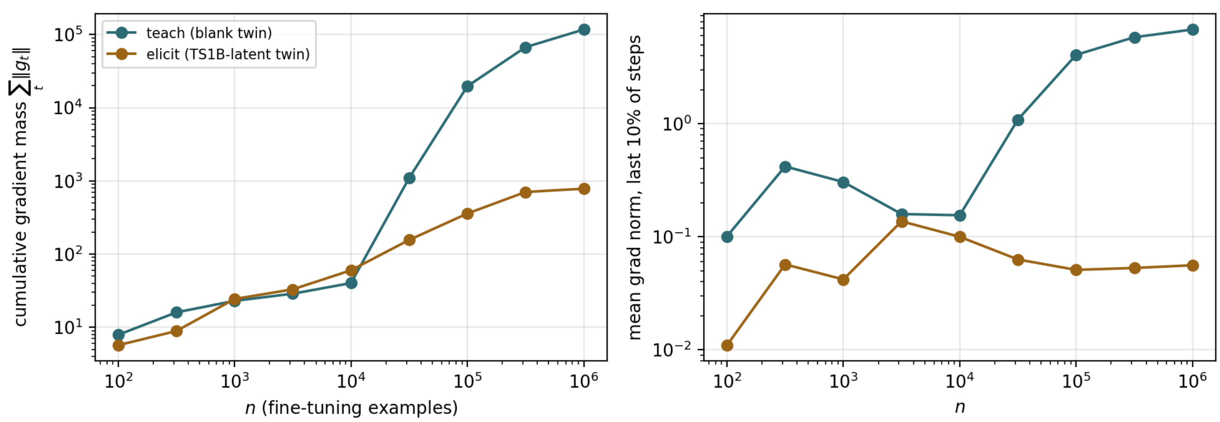

Figure 6: Cumulative gradient mass (left) and late-run gradient norm (right) vs n; teal = teach, gold = elicit.

Validation results (not regime differences, but what licenses R2–R6)

- **Both circuits are real and load-bearing.** True activation patching: elicited child top-8 of 528 nodes = 0.997 sufficient / 0.998 necessary (top-32: 0.999 / 1.000); taught child top-8 necessity 0.986, top-32 $\sim$0.999. Every overlap above compares provably load-bearing sets.
- **Construction controls.** Identical doses on the blank twin change nothing on the target (0.000 in every probe); a sum-only binding dose leaves "difference" unbound (per-word binding); the composition law chain = parse $\times$ compute held on four variants.

Informative non-discriminators (measured, reported for completeness)

- Effective rank of ΔW (participation ratio): elicit 7.1, teach 5.2 of 512 — both energy-concentrated; rank does not separate the regimes. Alignment of ΔW with the base's top-64 singular directions is near the 0.058 baseline for both.
- Temporal profile of weight travel: burst-then-decay in both arms (R7) — explained by AdamW normalisation; the underlying gradient dynamics do discriminate (R8).
- Prompt brittleness (16-shot collapse) is a property of this lineage's training-sequence structure, not of the regime: both children score $\sim$0 at 16 shots.
- Edge-level ΔS within the elicit arm: the edge instrument's split-half noise floor (0.61–0.62) exceeds the measured churn on both eval surfaces — a sensitivity limit.